# Bayesian Optimization Report: Function 4

## GP Health Summary per Iteration

| Iteration | GP Health | Status |
|---|---|---|
| 1 | 0.617 | Medium |
| 2 | 0.619 | Medium |
| 3 | 0.622 | Medium |
| 4 | 0.624 | Medium |
| 5 | 0.625 | Medium |
| 6 | 0.627 | Medium |
| 7 | 0.629 | Medium |
| 8 | 0.631 | Medium |
| 9 | 0.633 | Medium |
| 10 | 0.634 | Medium |

## GP Sigma Summary

| Iteration | Mean Sigma | Min Sigma | Max Sigma |
|---|---|---|---|
| 1 | 1.6279 | 0.4185 | 3.8521 |
| 2 | 1.5812 | 0.5975 | 3.9655 |
| 3 | 1.5501 | 0.7381 | 3.4716 |
| 4 | 1.4630 | 0.5624 | 4.2259 |
| 5 | 1.4364 | 0.7012 | 3.6665 |
| 6 | 1.4604 | 0.5727 | 4.4586 |
| 7 | 1.4298 | 0.4013 | 3.6810 |
| 8 | 1.3678 | 0.5109 | 3.2879 |
| 9 | 1.3280 | 0.5372 | 3.5360 |
| 10 | 1.3612 | 0.5487 | 3.3904 |

## Best Results Per Iteration

| Iteration | Best Input | Best Output | Kernel | Acquisition | GP Health | Anomalies |
|---|---|---|---|---|---|---|
| 1 | [1. 0.80765099 0.86224114 1. ] | -43.6443233996 3481 | Matern(ν=0.5) | EI | 0.617 | None |

| | | | | | | |
|---|---|---|---|---|---|---|
| 2 | [1. 0.82755924 0.98053371 0.9500518 ] | -45.8824207634779 | Matern(ν=0.5) | EI | 0.619 | None |
| 3 | [0.97733676 0.9910808 0.66762897 0.97332556] | -42.60753920861701 | Matern(ν=0.5) | EI | 0.622 | None |
| 4 | [1. 0.42891627 0.87219738 1. ] | -34.80751975548477 | Matern(ν=0.5) | EI | 0.624 | None |
| 5 | [1. 0.74735887 1. 1. ] | -45.392629858838845 | Matern(ν=0.5) | EI | 0.625 | None |
| 6 | [1. 0.92120991 0.92324764 1. ] | -48.33382592385269 | Matern(ν=0.5) | EI | 0.627 | None |
| 7 | [1. 0.99211713 1. 0.84769781] | -47.917066118891555 | Matern(ν=0.5) | EI | 0.629 | None |
| 8 | [1. 0.83003791 0.87373381 0.94076116] | -42.87536530757693 | Matern(ν=0.5) | EI | 0.631 | None |
| 9 | [0.55544196 1. 0.89271657 1. ] | -40.2071350194812 | Matern(ν=0.5) | EI | 0.633 | None |
| 10 | [1. 0.46361266 0.88636525 0.9714829 ] | -34.905288332667354 | Matern(ν=0.5) | EI | 0.634 | None |

## Acquisition Values Per Iteration

### *Iteration 1*

| Method | Max Value |
|---|---|
| UCB | 1.110008 |
| EI | 0.019397 |
| PI | 0.061760 |
| THOMPSON | 0.322230 |
| KG | 0.019397 |
| ENTROPY | inf |
| PORTFOLIO | 2.095000 |

### *Iteration 2*

| Method | Max Value |
|---|---|
| UCB | -0.066710 |
| EI | 0.003306 |
| PI | 0.011463 |
| THOMPSON | -0.266326 |
| KG | 0.003306 |
| ENTROPY | inf |

| | |
|---|---|
| PORTFOLIO | 1.552006 |

### *Iteration 3*

| Method | Max Value |
|---|---|
| UCB | -0.170034 |
| EI | 0.001344 |
| PI | 0.005448 |
| THOMPSON | -0.042951 |
| KG | 0.001344 |
| ENTROPY | 406.615132 |
| PORTFOLIO | 1.142826 |

### *Iteration 4*

| Method | Max Value |
|---|---|
| UCB | 0.213638 |
| EI | 0.008082 |
| PI | 0.032487 |
| THOMPSON | -0.224225 |
| KG | 0.008082 |
| ENTROPY | inf |
| PORTFOLIO | 1.127340 |

### *Iteration 5*

| Method | Max Value |
|---|---|
| UCB | 0.045502 |
| EI | 0.004849 |
| PI | 0.017343 |
| THOMPSON | -0.183542 |
| KG | 0.004849 |
| ENTROPY | inf |
| PORTFOLIO | 1.145289 |

### *Iteration 6*

| Method | Max Value |
|---|---|
| UCB | -0.122456 |

| | |
|---|---|
| EI | 0.001955 |
| PI | 0.007555 |
| THOMPSON | -0.870613 |
| KG | 0.001955 |
| ENTROPY | inf |
| PORTFOLIO | 1.010086 |

### *Iteration 7*

| Method | Max Value |
|---|---|
| UCB | 0.192523 |
| EI | 0.002678 |
| PI | 0.017422 |
| THOMPSON | -0.122258 |
| KG | 0.002678 |
| ENTROPY | inf |
| PORTFOLIO | 0.769865 |

### *Iteration 8*

| Method | Max Value |
|---|---|
| UCB | -0.125087 |
| EI | 0.002133 |
| PI | 0.008011 |
| THOMPSON | -0.737912 |
| KG | 0.002133 |
| ENTROPY | inf |
| PORTFOLIO | 0.619965 |

### *Iteration 9*

| Method | Max Value |
|---|---|
| UCB | 0.459293 |
| EI | 0.030855 |
| PI | 0.107522 |
| THOMPSON | 0.395691 |
| KG | 0.030855 |
| ENTROPY | inf |

| | |
|---|---|
| PORTFOLIO | 1.015905 |

### *Iteration 10*

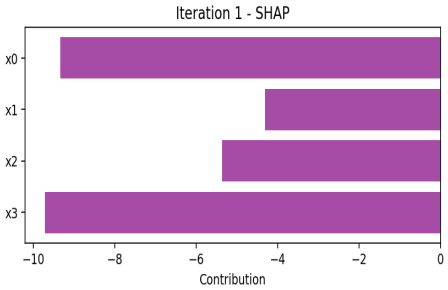
| Method | Max Value |
|---|---|
| UCB | 0.255370 |
| EI | 0.010553 |
| PI | 0.040672 |
| THOMPSON | 0.466794 |
| KG | 0.010553 |
| ENTROPY | inf |
| PORTFOLIO | 0.453912 |

## LOO MSE Summary

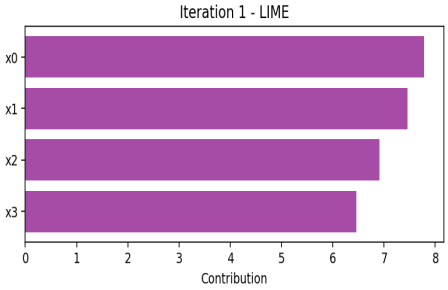
| Iteration | LOO MSE |
|---|---|
| 1 | 4.830085 |
| 2 | 3.861362 |
| 3 | 1.997051 |
| 4 | 1.894813 |
| 5 | 1.749928 |
| 6 | 1.730586 |
| 7 | 1.618786 |
| 8 | 2.887540 |
| 9 | 2.808089 |
| 10 | 2.525324 |

## Feature Explanations (SHAP / LIME)

### Iteration 1 - SHAP



## Iteration 1 - LIME



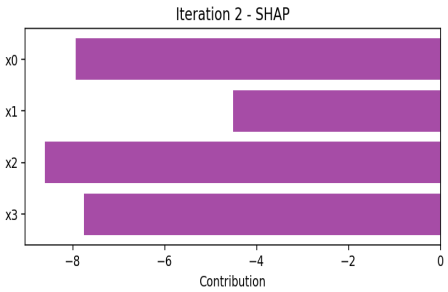

## Iteration 1 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -9.34074 |
| x1 | -4.31236 |
| x2 | -5.36525 |
| x3 | -9.71309 |

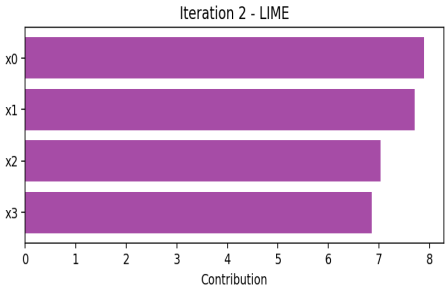
## Iteration 1 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 7.77839 |
| x1 | 7.45628 |
| x2 | 6.90985 |
| x3 | 6.46024 |

## Iteration 2 - SHAP




## Iteration 2 - LIME



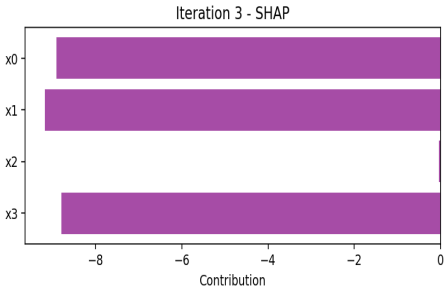

## Iteration 2 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -7.93544 |
| x1 | -4.51186 |
| x2 | -8.60064 |
| x3 | -7.74949 |

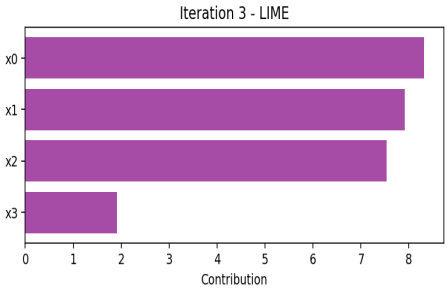
## Iteration 2 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 7.89059 |
| x1 | 7.70813 |
| x2 | 7.03458 |
| x3 | 6.85659 |

## Iteration 3 - SHAP




## Iteration 3 - LIME



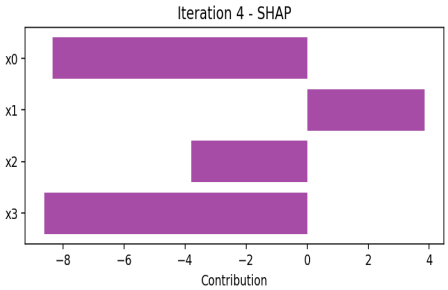

## Iteration 3 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -8.90756 |
| x1 | -9.17408 |
| x2 | -0.02955 |
| x3 | -8.79063 |

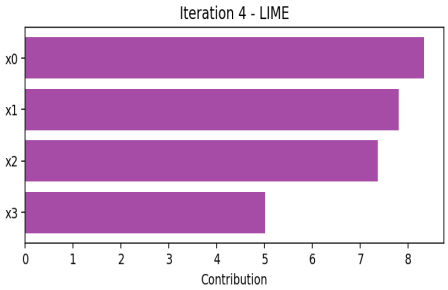
## Iteration 3 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.31563 |
| x1 | 7.92104 |
| x2 | 7.53568 |
| x3 | 1.91430 |

## Iteration 4 - SHAP




## Iteration 4 - LIME



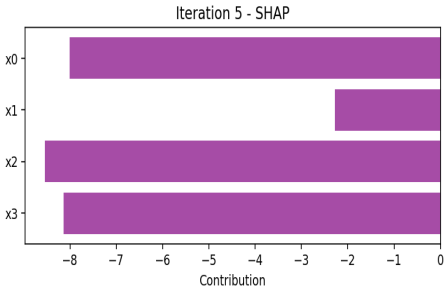

## Iteration 4 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -8.33812 |
| x1 | 3.85412 |
| x2 | -3.79846 |
| x3 | -8.61223 |

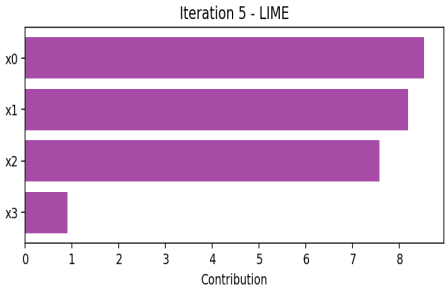
## Iteration 4 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.32848 |
| x1 | 7.80468 |
| x2 | 7.36737 |
| x3 | 5.00727 |

## Iteration 5 - SHAP




## Iteration 5 - LIME



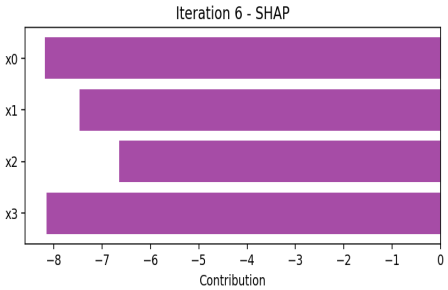

## Iteration 5 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -8.00839 |
| x1 | -2.28150 |
| x2 | -8.53744 |
| x3 | -8.13396 |

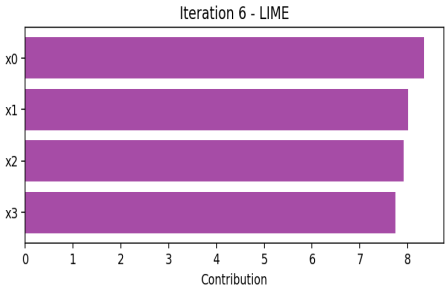
## Iteration 5 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.52128 |
| x1 | 8.18704 |
| x2 | 7.57519 |
| x3 | 0.90436 |

## Iteration 6 - SHAP




## Iteration 6 - LIME



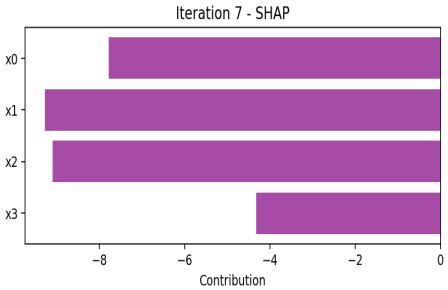

## Iteration 6 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -8.18324 |
| x1 | -7.46800 |
| x2 | -6.64656 |
| x3 | -8.15836 |

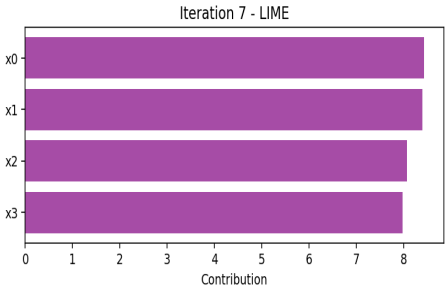
## Iteration 6 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.34274 |
| x1 | 8.01088 |
| x2 | 7.91483 |
| x3 | 7.74489 |

## Iteration 7 - SHAP




## Iteration 7 - LIME



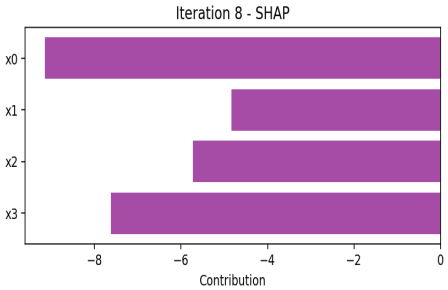

## Iteration 7 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -7.78170 |
| x1 | -9.27288 |
| x2 | -9.09873 |
| x3 | -4.32482 |

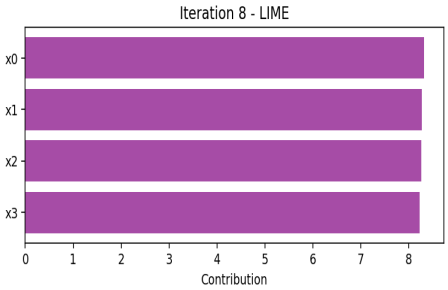
## Iteration 7 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.43302 |
| x1 | 8.40594 |
| x2 | 8.07492 |
| x3 | 7.98421 |

## Iteration 8 - SHAP




## Iteration 8 - LIME



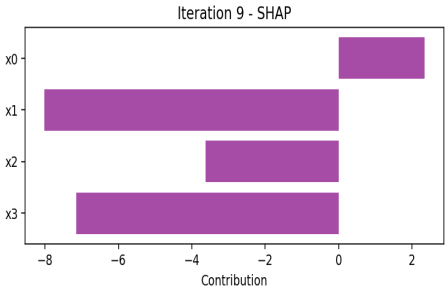

## Iteration 8 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -9.14435 |
| x1 | -4.82840 |
| x2 | -5.72411 |
| x3 | -7.62233 |

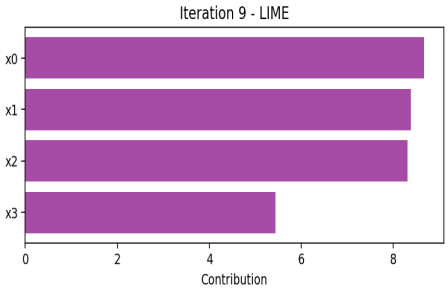
## Iteration 8 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.31793 |
| x1 | 8.28095 |
| x2 | 8.26882 |
| x3 | 8.22769 |

## Iteration 9 - SHAP




## Iteration 9 - LIME



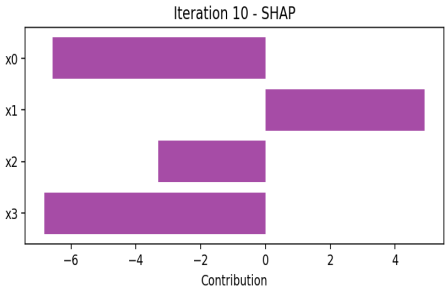

## Iteration 9 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | 2.35127 |
| x1 | -8.02058 |
| x2 | -3.62539 |
| x3 | -7.15646 |

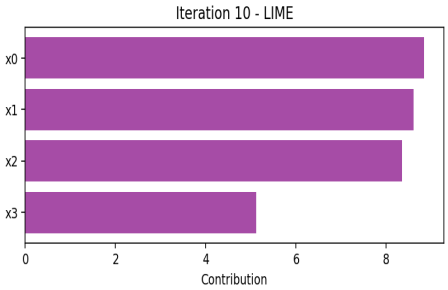
## Iteration 9 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.66246 |
| x1 | 8.37400 |
| x2 | 8.30948 |
| x3 | 5.44136 |

## Iteration 10 - SHAP




## Iteration 10 - LIME




## Iteration 10 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -6.56844 |
| x1 | 4.91914 |
| x2 | -3.30509 |
| x3 | -6.82620 |

## Iteration 10 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 8.83476 |
| x1 | 8.60135 |
| x2 | 8.35176 |
| x3 | 5.11640 |